

Creation Date 16-Jun-2009

Revision Date 20-Oct-2023

Revision Number 6

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                                  |                                   |
|----------------------------------|-----------------------------------|
| <b>Product Description:</b>      | <u><b>Acetonitrile</b></u>        |
| <b>Cat No. :</b>                 | <b>A956-1</b>                     |
| <b>Synonyms</b>                  | AN; Methyl cyanide; Ethanenitrile |
| <b>Index No</b>                  | 608-001-00-3                      |
| <b>CAS No</b>                    | 75-05-8                           |
| <b>EC No</b>                     | 200-835-2                         |
| <b>Molecular Formula</b>         | C <sub>2</sub> H <sub>3</sub> N   |
| <b>REACH registration number</b> | 01-2119471307-38-0053             |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Recommended Use</b>                | Laboratory chemicals. See Annex for full list.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Sector of use</b>                  | SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites<br>SU8 - Manufacture of bulk, large scale chemicals (including petroleum products)<br>SU9 - Manufacture of fine chemicals<br>SU10 - Formulation [mixing] of preparations and/or re-packaging (excluding alloys)<br>PC21 - Laboratory chemicals                                                                                                                                       |
| <b>Product category</b>               | see SECTION 16 for a complete list of uses for which an exposure scenario is provided as an annex                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Process categories</b>             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Environmental release category</b> | ERC1 - Manufacture of substances<br>ERC2 - Formulation of preparations (mixtures)<br>ERC4 - Industrial use of processing aids in processes and products, not becoming part of articles<br>ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates)<br>ERC6b - Industrial use of reactive processing aids<br>ERC7 - Industrial use of substances in closed systems<br>ERC8a - Wide dispersive indoor use of processing aids in open systems |
| <b>Uses advised against</b>           | SU21 - Consumer uses: Private households (= general public = consumers)<br>REACH Annex XVII Restriction - refer to SECTION 15                                                                                                                                                                                                                                                                                                                                                 |

### 1.3. Details of the supplier of the safety data sheet

|                       |                                                                                                                                                         |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Company</b>        | <b>EU entity/business name</b><br>Thermo Fisher Scientific<br>Janssen Pharmaceuticaaan 3a<br>2440 Geel, Belgium                                         |
|                       | <b>UK entity/business name</b><br>Fisher Scientific UK<br>Bishop Meadow Road, Loughborough,<br>Leicestershire LE11 5RG, United Kingdom                  |
|                       | <b>Swiss distributor</b> - Fisher Scientific AG<br>Neuhoferstrasse 11, CH 4153 Reinach<br>Tel: +41 (0) 56 618 41 11<br>e-mail - infoch@thermofisher.com |
| <b>E-mail address</b> | begel.sdsdesk@thermofisher.com                                                                                                                          |

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## 1.4. Emergency telephone number

Tel: 01509 231166  
Chemtrec US: (800) 424-9300  
Chemtrec EU: 001-703-527-3887

For customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

#### CLP Classification - Regulation (EC) No 1272/2008

##### Physical hazards

Flammable liquids Category 2 (H225)

##### Health hazards

Acute oral toxicity Category 4 (H302)  
Acute dermal toxicity Category 4 (H312)  
Acute Inhalation Toxicity - Vapors Category 4 (H332)  
Serious Eye Damage/Eye Irritation Category 2 (H319)

##### Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

### 2.2. Label elements



Signal Word

**Danger**

#### **Hazard Statements**

H225 - Highly flammable liquid and vapor  
H302 + H312 + H332 - Harmful if swallowed, in contact with skin or if inhaled  
H319 - Causes serious eye irritation

#### **Precautionary Statements**

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P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P301 + P312 - IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

## 2.3. Other hazards

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

Toxicity to Soil Dwelling Organisms

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component    | CAS No  | EC No     | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                               |
|--------------|---------|-----------|----------|-----------------------------------------------------------------------------------------------------------------|
| Acetonitrile | 75-05-8 | 200-835-2 | >95      | Flam. Liq. 2 (H225)<br>Acute Tox. 4 (H302)<br>Acute Tox. 4 (H312)<br>Eye Irrit. 2 (H319)<br>Acute Tox. 4 (H332) |

| Component    | ECHA (RAC) ATE (Oral) | ECHA (RAC) ATE (Dermal) | ECHA (RAC) ATE (Inhalation) |
|--------------|-----------------------|-------------------------|-----------------------------|
| Acetonitrile | ATE = 617 mg/kg       | -                       | -                           |

|                           |                       |
|---------------------------|-----------------------|
| REACH registration number | 01-2119471307-38-0053 |
|---------------------------|-----------------------|

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                       |                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b> | Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.                                                                                                                                                                                                                                                        |
| <b>Eye Contact</b>    | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                        |
| <b>Skin Contact</b>   | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                              |
| <b>Ingestion</b>      | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                                           |
| <b>Inhalation</b>     | Remove to fresh air. If breathing is irregular or stopped, administer artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. |

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**Self-Protection of the First Aider** Remove all sources of ignition. Use personal protective equipment as required. Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

## **4.2. Most important symptoms and effects, both acute and delayed**

Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Metabolism may release cyanide, which may result in headache, dizziness, weakness, collapse, unconsciousness, and possible death: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

## **4.3. Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically. The effects may be delayed therefore medical observation is essential. Effects may be delayed 7 to 10 hours. May be metabolized to cyanide which in turn acts by inhibiting cytochrome oxidase impairing cellular respiration.

## **SECTION 5: FIREFIGHTING MEASURES**

### **5.1. Extinguishing media**

#### **Suitable Extinguishing Media**

Water spray. CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

#### **Extinguishing media which must not be used for safety reasons**

Do not use a solid water stream as it may scatter and spread fire.

### **5.2. Special hazards arising from the substance or mixture**

Flammable. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Vapors may form explosive mixtures with air.

#### **Hazardous Combustion Products**

Hydrogen cyanide (hydrocyanic acid), Nitrogen oxides (NO<sub>x</sub>), Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### **5.3. Advice for firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## **SECTION 6: ACCIDENTAL RELEASE MEASURES**

### **6.1. Personal precautions, protective equipment and emergency procedures**

Remove all sources of ignition. Take precautionary measures against static discharges. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Ensure adequate ventilation. Use personal protective equipment as required.

### **6.2. Environmental precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information.

### **6.3. Methods and material for containment and cleaning up**

Remove all sources of ignition. Take precautionary measures against static discharges. Provide adequate ventilation. Use spark-proof tools and explosion-proof equipment. Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Prevent product from entering drains.

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## 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Keep away from open flames, hot surfaces and sources of ignition. Take precautionary measures against static discharges. Do not get in eyes, on skin, or on clothing. Do not breathe mist/vapors/spray. Use spark-proof tools and explosion-proof equipment. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded.

### Hygiene Measures

When using do not eat, drink or smoke. Provide regular cleaning of equipment, work area and clothing.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame. Flammables area.

Technical Rules for Hazardous Substances (TRGS) 510 Class 3  
Storage Class (LGK) (Germany)

### 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component    | European Union                                               | The United Kingdom                                                                                              | France                                                                                                                                                                 | Belgium                                                        | Spain                                                                                  |
|--------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Acetonitrile | TWA: 40 ppm (8hr)<br>TWA: 70 mg/m <sup>3</sup> (8hr)<br>Skin | STEL: 60 ppm 15 min<br>STEL: 102 mg/m <sup>3</sup> 15 min<br>TWA: 40 ppm 8 hr<br>TWA: 68 mg/m <sup>3</sup> 8 hr | TWA / VME: 40 ppm (8 heures). restrictive limit<br>TWA / VME: 70 mg/m <sup>3</sup> (8 heures). restrictive limit<br>TWA / VME: 5 mg/m <sup>3</sup> (8 heures).<br>Peau | TWA: 20 ppm 8 uren<br>TWA: 34 mg/m <sup>3</sup> 8 uren<br>Huid | TWA / VLA-ED: 40 ppm (8 horas)<br>TWA / VLA-ED: 68 mg/m <sup>3</sup> (8 horas)<br>Piel |

| Component    | Italy                                                                                                             | Germany                                                                                                                                                                                                    | Portugal                                                         | The Netherlands                  | Finland                                                                                                                                           |
|--------------|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Acetonitrile | TWA: 20 ppm 8 ore.<br>Time Weighted Average<br>TWA: 35 mg/m <sup>3</sup> 8 ore.<br>Time Weighted Average<br>Pelle | TWA: 10 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 17 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 10 ppm (8 Stunden). MAK<br>TWA: 17 mg/m <sup>3</sup> (8 Stunden). MAK<br>TWA: 2 | TWA: 40 ppm 8 horas<br>TWA: 70 mg/m <sup>3</sup> 8 horas<br>Pele | TWA: 34 mg/m <sup>3</sup> 8 uren | TWA: 20 ppm 8 tunteina<br>TWA: 34 mg/m <sup>3</sup> 8 tunteina<br>STEL: 40 ppm 15 minuutteina<br>STEL: 68 mg/m <sup>3</sup> 15 minuutteina<br>Iho |

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|  |  |                                                                                                                                         |  |  |  |
|--|--|-----------------------------------------------------------------------------------------------------------------------------------------|--|--|--|
|  |  | mg/m <sup>3</sup> (8 Stunden).<br>MAK<br>Höhepunkt: 20 ppm<br>Höhepunkt: 34 mg/m <sup>3</sup><br>Höhepunkt: 2 mg/m <sup>3</sup><br>Haut |  |  |  |
|--|--|-----------------------------------------------------------------------------------------------------------------------------------------|--|--|--|

| Component    | Austria                                                                                                                                                                | Denmark                                                                                                                                      | Switzerland                                                                                                                                               | Poland                                                                                 | Norway                                                                                                                                                                                                                    |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acetonitrile | Haut<br>MAK-KZGW: 160 ppm<br>15 Minuten<br>MAK-KZGW: 280 mg/m <sup>3</sup><br>15 Minuten<br>MAK-TMW: 40 ppm 8<br>Stunden<br>MAK-TMW: 70 mg/m <sup>3</sup> 8<br>Stunden | TWA: 40 ppm 8 timer<br>TWA: 70 mg/m <sup>3</sup> 8 timer<br>STEL: 80 ppm 15<br>minutter<br>STEL: 140 mg/m <sup>3</sup> 15<br>minutter<br>Hud | Haut/Peau<br>STEL: 40 ppm 15<br>Minuten<br>STEL: 68 mg/m <sup>3</sup> 15<br>Minuten<br>TWA: 20 ppm 8<br>Stunden<br>TWA: 34 mg/m <sup>3</sup> 8<br>Stunden | STEL: 140 mg/m <sup>3</sup> 15<br>minutach<br>TWA: 70 mg/m <sup>3</sup> 8<br>godzinach | TWA: 30 ppm 8 timer<br>TWA: 50 mg/m <sup>3</sup> 8 timer<br>TWA: 5 mg/m <sup>3</sup> 8 timer<br>STEL: 45 ppm 15<br>minutter. value<br>calculated<br>STEL: 75 mg/m <sup>3</sup> 15<br>minutter. value<br>calculated<br>Hud |

| Component    | Bulgaria                                                  | Croatia                                                                            | Ireland                                                                                                                       | Cyprus                                   | Czech Republic                                                                                                      |
|--------------|-----------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Acetonitrile | TWA: 40 ppm<br>TWA: 70 mg/m <sup>3</sup><br>Skin notation | kože<br>TWA-GVI: 40 ppm 8<br>satima.<br>TWA-GVI: 70 mg/m <sup>3</sup> 8<br>satima. | TWA: 40 ppm 8 hr.<br>TWA: 70 mg/m <sup>3</sup> 8 hr.<br>STEL: 120 ppm 15 min<br>STEL: 310 mg/m <sup>3</sup> 15<br>min<br>Skin | TWA: 40 ppm<br>TWA: 70 mg/m <sup>3</sup> | TWA: 70 mg/m <sup>3</sup> 8<br>hodinách.<br>Potential for cutaneous<br>absorption<br>Ceiling: 100 mg/m <sup>3</sup> |

| Component    | Estonia                                                                        | Gibraltar                                                           | Greece                                                                                  | Hungary                                                                                 | Iceland                                                                                                                                                |
|--------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acetonitrile | Nahk<br>TWA: 40 ppm 8<br>tundides.<br>TWA: 70 mg/m <sup>3</sup> 8<br>tundides. | Skin notation<br>TWA: 40 ppm 8 hr<br>TWA: 70 mg/m <sup>3</sup> 8 hr | STEL: 60 ppm<br>STEL: 105 mg/m <sup>3</sup><br>TWA: 40 ppm<br>TWA: 70 mg/m <sup>3</sup> | TWA: 70 mg/m <sup>3</sup> 8<br>óraban. AK<br>lehetséges borón<br>keresztüli felszívódás | TWA: 40 ppm 8<br>klukkustundum.<br>TWA: 70 mg/m <sup>3</sup> 8<br>klukkustundum.<br>Skin notation<br>Ceiling: 80 ppm<br>Ceiling: 140 mg/m <sup>3</sup> |

| Component    | Latvia                                                                                 | Lithuania                                                 | Luxembourg                                                                                                                  | Malta                                                                                             | Romania                                                               |
|--------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Acetonitrile | skin - potential for<br>cutaneous exposure<br>TWA: 40 ppm<br>TWA: 70 mg/m <sup>3</sup> | TWA: 40 ppm IPRD<br>TWA: 70 mg/m <sup>3</sup> IPRD<br>Oda | Possibility of significant<br>uptake through the skin<br>TWA: 40 ppm 8<br>Stunden<br>TWA: 70 mg/m <sup>3</sup> 8<br>Stunden | possibility of significant<br>uptake through the skin<br>TWA: 40 ppm<br>TWA: 70 mg/m <sup>3</sup> | Skin notation<br>TWA: 40 ppm 8 ore<br>TWA: 70 mg/m <sup>3</sup> 8 ore |

| Component    | Russia                    | Slovak Republic                                                                   | Slovenia                                                                                                                                  | Sweden                                                                                                                                                                             | Turkey                                                         |
|--------------|---------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Acetonitrile | MAC: 10 mg/m <sup>3</sup> | Potential for cutaneous<br>absorption<br>TWA: 40 ppm<br>TWA: 70 mg/m <sup>3</sup> | TWA: 40 ppm 8 urah<br>TWA: 70 mg/m <sup>3</sup> 8 urah<br>Koža<br>STEL: 140 mg/m <sup>3</sup> 15<br>minutah<br>STEL: 80 ppm 15<br>minutah | Indicative STEL: 60 ppm<br>15 minuter<br>Indicative STEL: 100<br>mg/m <sup>3</sup> 15 minuter<br>TLV: 30 ppm 8 timmar.<br>NGV<br>TLV: 50 mg/m <sup>3</sup> 8<br>timmar. NGV<br>Hud | Deri<br>TWA: 40 ppm 8 saat<br>TWA: 70 mg/m <sup>3</sup> 8 saat |

## Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

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MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                     | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|-------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Acetonitrile<br>75-05-8 (>95) |                              |                                 |                                | DNEL = 32.2mg/kg bw/day           |

| Component                     | Acute effects local (Inhalation)        | Acute effects systemic (Inhalation)     | Chronic effects local (Inhalation)      | Chronic effects systemic (Inhalation)   |
|-------------------------------|-----------------------------------------|-----------------------------------------|-----------------------------------------|-----------------------------------------|
| Acetonitrile<br>75-05-8 (>95) | DNEL = 40.6 ppm (68 mg/m <sup>3</sup> ) | DNEL = 40.6 ppm (68 mg/m <sup>3</sup> ) | DNEL = 40.6 ppm (68 mg/m <sup>3</sup> ) | DNEL = 40.6 ppm (68 mg/m <sup>3</sup> ) |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                     | Fresh water   | Fresh water sediment         | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)       |
|-------------------------------|---------------|------------------------------|--------------------|------------------------------------|--------------------------|
| Acetonitrile<br>75-05-8 (>95) | PNEC = 10mg/L | PNEC = 7.53mg/kg sediment dw | PNEC = 10mg/L      | PNEC = 32mg/L                      | PNEC = 2.41mg/kg soil dw |

| Component                     | Marine water | Marine water sediment | Marine water Intermittent | Food chain | Air |
|-------------------------------|--------------|-----------------------|---------------------------|------------|-----|
| Acetonitrile<br>75-05-8 (>95) | PNEC = 1mg/L |                       |                           |            |     |

## 8.2. Exposure controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting/equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material  | Breakthrough time | Glove thickness | EU standard    | Glove comments                                                                 |
|-----------------|-------------------|-----------------|----------------|--------------------------------------------------------------------------------|
| Butyl rubber    | > 480 minutes     | 0.35 mm         | EN 374 Level 6 | As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |
| Neoprene gloves | < 60 minutes      | 0.45 mm         |                |                                                                                |

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use

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appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** low boiling organic solvent Type AX Brown conforming to EN371

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141

**Environmental exposure controls** No information available.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                                |                                               |                                          |
|------------------------------------------------|-----------------------------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                                        |                                          |
| <b>Appearance</b>                              | Colorless                                     |                                          |
| <b>Odor</b>                                    | aromatic                                      |                                          |
| <b>Odor Threshold</b>                          | 170 ppm                                       |                                          |
| <b>Melting Point/Range</b>                     | -46 °C / -50.8 °F                             |                                          |
| <b>Softening Point</b>                         | No data available                             |                                          |
| <b>Boiling Point/Range</b>                     | 81 - 82 °C / 177.8 - 179.6 °F                 | @ 760 mmHg                               |
| <b>Flammability (liquid)</b>                   | Highly flammable                              | On basis of test data                    |
| <b>Flammability (solid,gas)</b>                | Not applicable                                | Liquid                                   |
| <b>Explosion Limits</b>                        | <b>Lower</b> 3 vol %<br><b>Upper</b> 16 vol % |                                          |
| <b>Flash Point</b>                             | 12.8 °C / 55 °F                               | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | 525 °C / 977 °F                               |                                          |
| <b>Decomposition Temperature</b>               | No data available                             |                                          |
| <b>pH</b>                                      | No information available                      |                                          |
| <b>Viscosity</b>                               | 0.36 cP at 20 °C                              |                                          |
| <b>Water Solubility</b>                        | Miscible                                      |                                          |
| <b>Solubility in other solvents</b>            | No information available                      |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                               |                                          |
| <b>Component</b>                               | <b>log Pow</b>                                |                                          |
| Acetonitrile                                   | -0.34                                         |                                          |
| <b>Vapor Pressure</b>                          | 97 mbar @ 20 °C                               |                                          |
| <b>Density / Specific Gravity</b>              | 0.781                                         |                                          |
| <b>Bulk Density</b>                            | Not applicable                                | Liquid                                   |
| <b>Vapor Density</b>                           | 1.42                                          | (Air = 1.0)                              |
| <b>Particle characteristics</b>                | Not applicable (liquid)                       |                                          |

### 9.2. Other information

|                             |                                                           |
|-----------------------------|-----------------------------------------------------------|
| <b>Molecular Formula</b>    | C2 H3 N                                                   |
| <b>Molecular Weight</b>     | 41.05                                                     |
| <b>Explosive Properties</b> | Not explosive Vapors may form explosive mixtures with air |
| <b>Oxidizing Properties</b> | Not oxidising                                             |
| <b>Evaporation Rate</b>     | 5.79 - (Butyl Acetate = 1.0)                              |

## SECTION 10: STABILITY AND REACTIVITY



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## 10.1. Reactivity

None known, based on information available

## 10.2. Chemical stability

Stable under normal conditions.

## 10.3. Possibility of hazardous reactions

### Hazardous Polymerization Hazardous Reactions

Hazardous polymerization does not occur.  
No information available.

## 10.4. Conditions to avoid

Incompatible products. Keep away from open flames, hot surfaces and sources of ignition.  
Exposure to moisture.

## 10.5. Incompatible materials

Strong oxidizing agents. Strong acids. Reducing Agent. Bases.

## 10.6. Hazardous decomposition products

Hydrogen cyanide (hydrocyanic acid). Nitrogen oxides (NO<sub>x</sub>). Carbon monoxide (CO).  
Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

|            |            |
|------------|------------|
| Oral       | Category 4 |
| Dermal     | Category 4 |
| Inhalation | Category 4 |

| Component    | LD50 Oral                                 | LD50 Dermal             | LC50 Inhalation                                                                         |
|--------------|-------------------------------------------|-------------------------|-----------------------------------------------------------------------------------------|
| Acetonitrile | 450-787 mg/kg (Rat)<br>2460 mg/kg ( Rat ) | > 2000 mg/kg ( Rabbit ) | LC50 = 3587 ppm (6.022 mg/l)<br>(Mouse) 4h<br>LC50 = 16,000 ppm (26.8 mg/l)<br>(Rat) 4h |

| Component    | ECHA (RAC) ATE (Oral) | ECHA (RAC) ATE (Dermal) | ECHA (RAC) ATE (Inhalation) |
|--------------|-----------------------|-------------------------|-----------------------------|
| Acetonitrile | ATE = 617 mg/kg       | -                       | -                           |

(b) skin corrosion/irritation; Based on available data, the classification criteria are not met

(c) serious eye damage/irritation; Category 2

#### (d) respiratory or skin sensitization;

|             |                                                                  |
|-------------|------------------------------------------------------------------|
| Respiratory | Based on available data, the classification criteria are not met |
| Skin        | Based on available data, the classification criteria are not met |

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

(f) carcinogenicity; Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product

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(g) reproductive toxicity; Based on available data, the classification criteria are not met

(h) STOT-single exposure; Based on available data, the classification criteria are not met

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

Target Organs None known.

(j) aspiration hazard; Based on available data, the classification criteria are not met

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Metabolism may release cyanide, which may result in headache, dizziness, weakness, collapse, unconsciousness, and possible death. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

## 11.2. Information on other hazards

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

#### Ecotoxicity effects

| Component    | Freshwater Fish                                                                                                                                                                                                                     | Water Flea | Freshwater Algae |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|
| Acetonitrile | LC50: = 1850 mg/L, 96h static (Lepomis macrochirus)<br>LC50: = 1000 mg/L, 96h static (Pimephales promelas)<br>LC50: 1600 - 1690 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: = 1650 mg/L, 96h static (Poecilia reticulata) |            |                  |

| Component    | Microtox                                                               | M-Factor |
|--------------|------------------------------------------------------------------------|----------|
| Acetonitrile | EC50 = 28000 mg/L 48 h<br>EC50 = 73 mg/L 24 h<br>EC50 = 7500 mg/L 15 h |          |

### 12.2. Persistence and degradability

**Persistence** Persistence is unlikely, based on information available.

**12.3. Bioaccumulative potential** Bioaccumulation is unlikely

| Component    | log Pow | Bioconcentration factor (BCF) |
|--------------|---------|-------------------------------|
| Acetonitrile | -0.34   | No data available             |

### 12.4. Mobility in soil

The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air.

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## 12.5. Results of PBT and vPvB assessment

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

## 12.6. Endocrine disrupting properties

### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

## 12.7. Other adverse effects

### Persistent Organic Pollutant

### Ozone Depletion Potential

This product does not contain any known or suspected substance

This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

#### Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

#### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

#### European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

#### Other Information

Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations.

#### Switzerland - Waste Ordinance

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

#### 14.1. UN number

UN1648

#### 14.2. UN proper shipping name

ACETONITRILE

#### 14.3. Transport hazard class(es)

3

#### 14.4. Packing group

II

### ADR

#### 14.1. UN number

UN1648

#### 14.2. UN proper shipping name

ACETONITRILE

#### 14.3. Transport hazard class(es)

3

#### 14.4. Packing group

II

### IATA

#### 14.1. UN number

UN1648

#### 14.2. UN proper shipping name

ACETONITRILE

#### 14.3. Transport hazard class(es)

3

#### 14.4. Packing group

II

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- 14.5. Environmental hazards** No hazards identified
- 14.6. Special precautions for user** No special precautions required.
- 14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component    | CAS No  | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|--------------|---------|-----------|--------|-----|-------|------|----------|------|------|
| Acetonitrile | 75-05-8 | 200-835-2 | -      | -   | X     | X    | KE-00067 | X    | X    |

| Component    | CAS No  | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDL | AICS | NZIoC | PICCS |
|--------------|---------|------|-----------------------------------------------|-----|-----|------|-------|-------|
| Acetonitrile | 75-05-8 | X    | ACTIVE                                        | X   | -   | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

#### Authorisation/Restrictions according to EU REACH

| Component    | CAS No  | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|--------------|---------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Acetonitrile | 75-05-8 | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

#### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

#### Seveso III Directive (2012/18/EC)

| Component    | CAS No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|--------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Acetonitrile | 75-05-8 | Not applicable                                                                            | Not applicable                                                                           |

#### Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

#### Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

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Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

**WGK Classification** See table for values

| Component    | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|--------------|---------------------------------------|-------------------------|
| Acetonitrile | WGK2                                  |                         |

| Component    | France - INRS (Tables of occupational diseases)      |
|--------------|------------------------------------------------------|
| Acetonitrile | Tableaux des maladies professionnelles (TMP) - RG 84 |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has been conducted by the manufacturer/importer

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H225 - Highly flammable liquid and vapor

H302 - Harmful if swallowed

H312 - Harmful in contact with skin

H319 - Causes serious eye irritation

H332 - Harmful if inhaled

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

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Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts. Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

|                  |                 |
|------------------|-----------------|
| Creation Date    | 16-Jun-2009     |
| Revision Date    | 20-Oct-2023     |
| Revision Summary | Not applicable. |

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and  
Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**

## Annex to the Safety Data Sheet according to Regulation (EC) No 1907/2006 [REACH]

### Acetonitrile - Exposure Scenarios

|                          |                                                           |                           |
|--------------------------|-----------------------------------------------------------|---------------------------|
| <b>CAS No</b><br>75-05-8 | <b>REACH registration number</b><br>01-2119471307-38-xxxx | <b>EC No</b><br>203-726-8 |
|--------------------------|-----------------------------------------------------------|---------------------------|

| Exposure Scenarios Overview                                                         |                                                                                                                                                                                                                       |                        |                                                                                                                                                                                                            |                     |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Title                                                                               | Sector of use                                                                                                                                                                                                         | Process category(ies)  | Environmental release category                                                                                                                                                                             | ES Identifier       |
| Manufacture of acetonitrile                                                         | SU3 - Industrial uses:<br>Uses of substances as such or in preparations at industrial sites<br>SU8 - Manufacture of bulk, large scale chemicals (including petroleum products)<br>SU9 - Manufacture of fine chemicals | 1, 2, 3, 4, 8a, 8b, 9  | ERC1 - Manufacture of substances                                                                                                                                                                           | ES1-M1 ACETONITRILE |
| Industrial use of acetonitrile                                                      | SU3 - Industrial uses:<br>Uses of substances as such or in preparations at industrial sites<br>SU8 - Manufacture of bulk, large scale chemicals (including petroleum products)<br>SU9 - Manufacture of fine chemicals | 1, 2, 3, 4, 8a, 8b, 9  | ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates)<br>ERC6b - Industrial use of reactive processing aids<br>ERC7 - Industrial use of substances in closed systems | ES2-M2 ACETONITRILE |
| Pharmaceutical, fine chemical and active substance manufacture uses of acetonitrile | SU9 - Manufacture of fine chemicals                                                                                                                                                                                   | 1, 2, 3, 4, 8a, 8b, 15 | ERC4 - Industrial use of processing aids in processes and products, not becoming part of articles<br>ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates)           | ES3-M3 ACETONITRILE |
| Formulation of preparations and/or re-packaging                                     | SU10 - Formulation [mixing] of preparations and/or re-packaging (excluding alloys)                                                                                                                                    | 3, 5, 9                | ERC2 - Formulation of preparations                                                                                                                                                                         | ES4-F1 ACETONITRILE |
| Laboratory use                                                                      | SU3 - Industrial uses:<br>Uses of substances as such or in preparations at industrial sites                                                                                                                           | 3, 15                  | ERC8a - Wide dispersive indoor use of processing aids in open systems                                                                                                                                      | ES5-L1 ACETONITRILE |

### Exposure scenario

#### ES1 Manufacture of Acetonitrile - ES1-M1 ACETONITRILE

### Section 1 - Identification of the use

|                                             |                                                                                                                                                                                |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Main user group</b>                      | Industrial uses: Uses of substances as such or in preparations at industrial sites                                                                                             |
| <b>Type</b>                                 | Worker                                                                                                                                                                         |
| <b>Processes, tasks, activities covered</b> | Manufacture or use as an intermediate or process chemical or extraction agent. Loading (including marine vessel/barge, rail/road car and IBC loading) and repacking (including |

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | drums and small packs) of substance, including its sampling, storage, unloading distribution and associated laboratory activities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Sector(s) of use</b>                    | SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites<br>SU8 - Manufacture of bulk, large scale chemicals (including petroleum products)<br>SU9 - Manufacture of fine chemicals                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Process category(ies)</b>               | PROC1 - Use in closed process, no likelihood of exposure<br>PROC2 - Use in closed, continuous process with occasional controlled exposure<br>PROC3 - Use in closed batch process (synthesis or formulation)<br>PROC4 - Use in batch and other process (synthesis) where opportunity for exposure arises<br>PROC8a - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at non dedicated facilities<br>PROC8b - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities<br>PROC9 - Transfer of substance or preparation into small containers (dedicated filling line, including weighing) |
| <b>Environmental release category(ies)</b> | ERC1 - Manufacture of substances                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

## Section 2 - Operational Conditions and Risk Management Measures

### Product characteristics

|                                   |                          |
|-----------------------------------|--------------------------|
| <b>Physical State</b>             | Liquid                   |
| <b>pH</b>                         | No information available |
| <b>Water Solubility</b>           | Miscible                 |
| <b>Vapor Pressure</b>             | 97 mbar @ 20 °C          |
| <b>Volatility</b>                 | High                     |
| Covers concentrations up to 100 % |                          |

### Section 2.1 - Control of environmental exposure

#### Environmental release category(ies)

ERC1 - Manufacture of substances

#### Specific Environmental Release Category

ESVOC SpERC 1.1.v1

#### Control of environmental exposure

Readily biodegradable  
Annual amount used in the EU 8500 t/a  
Annual amount per site 1000 t/a (Worst case)

#### Environmental factors not influenced by risk management

|                                            |           |
|--------------------------------------------|-----------|
| Emission days                              | 300       |
| Receiving water dilution (fresh or marine) | 2000 m3/d |

#### Other operational conditions of use affecting environmental exposure

|                                                                                                                              |                                  |
|------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| Emission days                                                                                                                | 300 (from ESVOC SPERC 1.1.v1)    |
| Release fraction to wastewater from process (initial release prior to RMM)                                                   | 1% (Specified by ESVOC 1.1.v1)   |
| Release fraction to air from process (after typical onsite RMMs consistent with EU Solvent Emissions Directive requirements) | 0.5% (Specified by ESVOC 1.1.v1) |

#### Technical onsite conditions and measures to reduce or limit discharges, air emissions

Technical onsite conditions and measures to reduce or limit discharges, air emissions  
Negligible air emissions as process operates in a contained system.  
Additional good practice advice beyond the REACH Chemical Safety Report  
Bund storage facilities to prevent soil and water pollution in the event of spillage. Ensure all waste water is collected and treated via a WWTP.

#### Conditions and measures related to municipal sewage treatment plant

|         |                                                                                                                           |
|---------|---------------------------------------------------------------------------------------------------------------------------|
| Remarks | Manufacturing plants will have on-site waste water treatment facilities and emission to the municipal STP will not occur. |
|---------|---------------------------------------------------------------------------------------------------------------------------|

#### Waste management



|       |                                                        |
|-------|--------------------------------------------------------|
| Air   | 142 kg/day Based on ESVOG SPERC 1.1.v1 release factors |
| Water | 283 kg/d Based on ESVOG SPERC 1.1.v1 release factors   |
| Soil  | 0.01% ERC release factor                               |

#### Conditions and measures related to external treatment of waste for disposal

|                         |                                                                   |
|-------------------------|-------------------------------------------------------------------|
| Disposal                | Waste resulting from on-site RMM to be disposed as chemical waste |
| Waste treatment methods | Municipal waste incineration                                      |

## Section 2.2 - Control of worker exposure

#### General information on risk management related to physicochemical hazard

Remove all sources of ignition. Take precautionary measures against static charges. Use only non-sparking tools. Keep equipment under negative pressure. Segregate work area and mark with appropriate signs in accordance with local/regional/national legislation.

#### General information on exposure estimation

Manufactured and processed at industrial sites in closed continuous processes with either no likelihood of exposure or with only occasional opportunity for exposure in controlled conditions e.g. during maintenance, sampling or discharge of the material. Transfer of the substance is conducted at dedicated facilities using a closed-system with vapour return. Respiratory protection is not required except for certain critical activities where respiratory protective equipment is used, for example, cleaning tanks or reactors. Measured dermal exposure data are not available.

#### Control of worker exposure

|                                                                                        |                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Process category(ies)                                                                  | PROC1 - Use in closed process, no likelihood of exposure                                                                                                                                                                                               |
| Covers concentrations up to                                                            | 100%                                                                                                                                                                                                                                                   |
| Amounts used                                                                           | >1000 t/y                                                                                                                                                                                                                                              |
| Exposure duration                                                                      | Avoid carrying out operation for more than 8h                                                                                                                                                                                                          |
| Use frequency                                                                          | 220 days per year                                                                                                                                                                                                                                      |
| Indoor/Outdoor use                                                                     | Outdoor                                                                                                                                                                                                                                                |
| Assumes process temperature up to                                                      | <=40°C                                                                                                                                                                                                                                                 |
| Organisational measures to prevent /limit releases, dispersion and exposure            | Handle substance within a closed system Use of closed transfers of liquids from storage to production equipment (e.g. metered piped or pumped additions) Sample via a closed loop or other system to avoid exposure                                    |
| Technical conditions and measures to control dispersion from source towards the worker | Undertake operation under enclosed conditions                                                                                                                                                                                                          |
| Conditions and measures related to personal protection, hygiene and health evaluation  | Use eye protection according to EN 166, designed to protect against liquid splashes<br>Wear chemically resistant gloves (tested to EN374) in combination with specific activity training                                                               |
| Additional good practice advice beyond the REACH Chemical Safety Report                | Workers involved in production, handling, sampling and transfer of materials are well-trained in these procedures as well as good industrial hygiene practices<br>-----                                                                                |
| Process category(ies)                                                                  | PROC2 - Use in closed, continuous process with occasional controlled exposure                                                                                                                                                                          |
| Covers concentrations up to                                                            | 100%                                                                                                                                                                                                                                                   |
| Exposure duration                                                                      | Avoid carrying out operation for more than 8h                                                                                                                                                                                                          |
| Indoor/Outdoor use                                                                     | Outdoor                                                                                                                                                                                                                                                |
| Assumes process temperature up to                                                      | <=40°C                                                                                                                                                                                                                                                 |
| Conditions and measures related to personal protection, hygiene and health evaluation  | Wear a respirator providing a minimum efficiency of 90% (APF 10) Use eye protection according to EN 166, designed to protect against liquid splashes Wear chemically resistant gloves (tested to EN374) in combination with specific activity training |
| Additional good practice advice beyond the REACH Chemical Safety Report                | Workers involved in production, handling, sampling and transfer of materials are well-trained in these procedures as well as good industrial hygiene practices<br>-----                                                                                |
| Process category(ies)                                                                  | PROC3 - Use in closed batch process (synthesis or formulation)                                                                                                                                                                                         |
| Covers concentrations up to                                                            | 100%                                                                                                                                                                                                                                                   |
| Exposure duration                                                                      | < 1 hour(s)                                                                                                                                                                                                                                            |
| Indoor/Outdoor use                                                                     | Outdoor                                                                                                                                                                                                                                                |
| Assumes process temperature up to                                                      | <=40°C                                                                                                                                                                                                                                                 |
| Conditions and measures related to personal protection, hygiene and                    | Use eye protection according to EN 166, designed to protect against liquid splashes Wear chemically resistant gloves (tested to EN374) in combination with specific activity training                                                                  |

|                                                                                       |                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| health evaluation                                                                     | Wear a respirator providing a minimum efficiency of 90% (APF 10)                                                                                                                                                                                             |
| Additional good practice advice beyond the REACH Chemical Safety Report               | Workers involved in production, handling, sampling and transfer of materials are well-trained in these procedures as well as good industrial hygiene practices<br>-----                                                                                      |
| Process category(ies)                                                                 | PROC4 - Use in batch and other process (synthesis) where opportunity for exposure arises                                                                                                                                                                     |
| Covers concentrations up to                                                           | 100%                                                                                                                                                                                                                                                         |
| Exposure duration                                                                     | Avoid carrying out activities involving exposure for more than 1 hour                                                                                                                                                                                        |
| Indoor/Outdoor use                                                                    | Outdoor                                                                                                                                                                                                                                                      |
| Assumes process temperature up to                                                     | <=40°C                                                                                                                                                                                                                                                       |
| Conditions and measures related to personal protection, hygiene and health evaluation | Use eye protection according to EN 166, designed to protect against liquid splashes Wear a respirator providing a minimum efficiency of 90% (APF 10) Wear chemically resistant gloves (tested to EN374) in combination with specific activity training       |
| Additional good practice advice beyond the REACH Chemical Safety Report               | Workers involved in production, handling, sampling and transfer of materials are well-trained in these procedures as well as good industrial hygiene practices<br>-----                                                                                      |
| Process category(ies)                                                                 | PROC8a - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at non dedicated facilities                                                                                                                            |
| Covers concentrations up to                                                           | 100%                                                                                                                                                                                                                                                         |
| Exposure duration                                                                     | < 1 hour(s)                                                                                                                                                                                                                                                  |
| Indoor/Outdoor use                                                                    | Outdoor                                                                                                                                                                                                                                                      |
| Assumes process temperature up to                                                     | <=40°C                                                                                                                                                                                                                                                       |
| Organisational measures to prevent /limit releases, dispersion and exposure           | Avoid carrying out operation for more than 1 hour<br>Ensure operation is undertaken outdoors                                                                                                                                                                 |
| Conditions and measures related to personal protection, hygiene and health evaluation | Use eye protection according to EN 166, designed to protect against liquid splashes<br>Wear a respirator providing a minimum efficiency of 95% (APF 20)<br>Wear chemically resistant gloves (tested to EN374) in combination with specific activity training |
| Additional good practice advice beyond the REACH Chemical Safety Report               | Assumes a good basic standard of occupational hygiene is implemented<br>-----                                                                                                                                                                                |
| Process category(ies)                                                                 | PROC8b - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities                                                                                                                                |
| Covers concentrations up to                                                           | 100%                                                                                                                                                                                                                                                         |
| Exposure duration                                                                     | Avoid carrying out activities involving exposure for more than 1 hour                                                                                                                                                                                        |
| Indoor/Outdoor use                                                                    | Outdoor                                                                                                                                                                                                                                                      |
| Assumes process temperature up to                                                     | <=40°C                                                                                                                                                                                                                                                       |
| Conditions and measures related to personal protection, hygiene and health evaluation | Use eye protection according to EN 166, designed to protect against liquid splashes Wear chemically resistant gloves (tested to EN374) in combination with specific activity training                                                                        |
| Additional good practice advice beyond the REACH Chemical Safety Report               | Assumes a good basic standard of occupational hygiene is implemented<br>-----                                                                                                                                                                                |
| Process category(ies)                                                                 | PROC9 - Transfer of substance or preparation into small containers (dedicated filling line, including weighing)                                                                                                                                              |
| Covers concentrations up to                                                           | 100%                                                                                                                                                                                                                                                         |
| Exposure duration                                                                     | Avoid carrying out operation for more than 8h                                                                                                                                                                                                                |
| Indoor/Outdoor use                                                                    | Indoor use                                                                                                                                                                                                                                                   |
| Assumes process temperature up to                                                     | <=40°C                                                                                                                                                                                                                                                       |
| Minimum room ventilation rate for handling/application (air changes per hour)         | 1-3                                                                                                                                                                                                                                                          |
| Organisational measures to prevent /limit releases, dispersion and exposure           | Fill containers/cans at dedicated fill points supplied with local extract ventilation                                                                                                                                                                        |
| Conditions and measures related to personal protection, hygiene and health evaluation | Use eye protection according to EN 166, designed to protect against liquid splashes Wear chemically resistant gloves (tested to EN374) in combination with specific activity training<br>Wear a respirator providing a minimum efficiency of 90%<br>-----    |

Control of consumer exposure Not intended for consumer use

### Section 3 - Exposure estimation

#### Environment

##### Environmental release category(ies)

ERC1 - Manufacture of substances

Specific Environmental Release Category - ESVOC SpERC 1.1.v1

Predicted No Effect Concentration (PNEC) - See values below

|                                    |             |                       |               |
|------------------------------------|-------------|-----------------------|---------------|
| Fresh water                        | 10 mg/l     | Marine water          | 1 mg/l        |
| Fresh water sediment               | 45 mg/kg dw | Marine water sediment | 4.5 mg/kg dw  |
| Water Intermittent                 | 10 mg/l     | Soil (Agriculture)    | 2.41 mg/kg dw |
| Microorganisms in sewage treatment | 32 mg/l     |                       |               |

| Environment         | Predicted exposure level                | Risk characterization ratio (RCR) |
|---------------------|-----------------------------------------|-----------------------------------|
| Freshwater          | $2.22 \times 10^{-4}$ mg/l              | <0.01                             |
| Marine water        | $2.06 \times 10^{-5}$ mg/l              | <0.01                             |
| Freshwater sediment | $8.5 \times 10^{-4}$ mg/kg dw           | <0.01                             |
| Marine sediment     | $8.02 \times 10^{-5}$ mg/kg dw          | <0.01                             |
| Soil                | $4.62 \times 10^{-6}$ mg/kg dw          | <0.01                             |
| Air                 | $2.27 \times 10^{-6}$ mg/m <sup>3</sup> |                                   |

Calculation method - EUSES 2.1

#### Remarks

No significant PEC values are indicated for the regional scale even under the conservative assumptions of the Tier 2 EUSES assessment. All derived PECs are below the relevant PNEC and so no further assessment or refinements are required.

#### Health

Derived No Effect Level (DNEL) - See table for values

| Route of exposure | Acute effects (local)               | Acute effects (systemic)            | Chronic effects (local)             | Chronic effects (systemic)                               |
|-------------------|-------------------------------------|-------------------------------------|-------------------------------------|----------------------------------------------------------|
| Oral              |                                     |                                     |                                     |                                                          |
| Dermal            |                                     |                                     |                                     |                                                          |
| Inhalation        | 40.6 ppm<br>(68 mg/m <sup>3</sup> ) | 40.6 ppm<br>(68 mg/m <sup>3</sup> ) | 40.6 ppm<br>(68 mg/m <sup>3</sup> ) | 32.2 mg/kg bw/day<br>40.6 ppm<br>(68 mg/m <sup>3</sup> ) |

| Process category(ies)                                                                                                             | Exposure route      | Predicted exposure level | Risk characterization ratio (RCR) |
|-----------------------------------------------------------------------------------------------------------------------------------|---------------------|--------------------------|-----------------------------------|
| PROC1 - Use in closed process, no likelihood of exposure                                                                          | Worker - inhalative | 0.012 mg/m <sup>3</sup>  | <0.01                             |
|                                                                                                                                   | Worker - dermal     | 0.343 mg/kg bw/day       | 0.011                             |
| PROC2 - Use in closed, continuous process with occasional controlled exposure                                                     | Worker - inhalative | 12.0 mg/m <sup>3</sup>   | 0.179                             |
|                                                                                                                                   | Worker - dermal     | 1.37 mg/kg bw/day        | 0.043                             |
| PROC3 - Use in closed batch process (synthesis or formulation)                                                                    | Worker - inhalative | 29.9 mg/m <sup>3</sup>   | 0.447                             |
|                                                                                                                                   | Worker - dermal     | 0.343 mg/kg bw/day       | 0.011                             |
| PROC4 - Use in batch and other process (synthesis) where opportunity for exposure arises                                          | Worker - inhalative | 24.0 mg/m <sup>3</sup>   | 0.357                             |
|                                                                                                                                   | Worker - dermal     | 6.86 mg/kg bw/day        | 0.214                             |
| PROC8a - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at non dedicated facilities | Worker - inhalative | 60.0 mg/m <sup>3</sup>   | 0.894                             |
|                                                                                                                                   | Worker - dermal     | 12.0 mg/kg bw/day        | 0.429                             |

|                                                                                                                               |                     |                         |       |
|-------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------|-------|
| PROC8b - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities | Worker - inhalative | 60.0 mg/m <sup>3</sup>  | 0.894 |
|                                                                                                                               | Worker - dermal     | 6.86 mg/kg bw/day       | 0.214 |
| PROC9 - Transfer of substance or preparation into small containers (dedicated filling line, including weighing)               | Worker - inhalative | 0.855 mg/m <sup>3</sup> | 0.013 |
|                                                                                                                               | Worker - dermal     | 6.86 mg/kg bw/day       | 0.021 |

**Calculation method** Used ECETOC TRA model

#### Remarks

Predicted exposures are not expected to exceed the DN(M)EL when the risk management measures/operational conditions outlined in section 2 are implemented

### Section 4 - Guidance to check compliance with the exposure scenario

Used EUSES model

Used ECETOC TRA model

Further details on scaling and control technologies are provided in SpERC factsheet (<http://cefic.org/en/reach-for-industries-libraries.html>)

Predicted exposures are not expected to exceed the applicable exposure limits (given in section 8 of the SDS) when the operational conditions/risk management measures given in section 2 are implemented  
ECHA guidance for downstream users

## Annex to the Safety Data Sheet according to Regulation (EC) No 1907/2006 [REACH]

### Acetonitrile - Exposure Scenarios

|                   |                                                    |                    |
|-------------------|----------------------------------------------------|--------------------|
| CAS No<br>75-05-8 | REACH registration number<br>01-2119471307-38-xxxx | EC No<br>203-726-8 |
|-------------------|----------------------------------------------------|--------------------|

#### Exposure scenario

#### ES2 Industrial use of Acetonitrile - ES2-M2 ACETONITRILE

#### Section 1 - Identification of the use

|                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Main user group</b>                      | Industrial uses: Uses of substances as such or in preparations at industrial sites                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Type</b>                                 | Worker                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Processes, tasks, activities covered</b> | Manufacture or use as an intermediate or process chemical or extraction agent. Loading (including marine vessel/barge, rail/road car and IBC loading) and repacking (including drums and small packs) of substance, including its sampling, storage, unloading distribution and associated laboratory activities                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Sector(s) of use</b>                     | SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites<br>SU8 - Manufacture of bulk, large scale chemicals (including petroleum products)<br>SU9 - Manufacture of fine chemicals                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Process category(ies)</b>                | PROC1 - Use in closed process, no likelihood of exposure<br>PROC2 - Use in closed, continuous process with occasional controlled exposure<br>PROC3 - Use in closed batch process (synthesis or formulation)<br>PROC4 - Use in batch and other process (synthesis) where opportunity for exposure arises<br>PROC8a - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at non dedicated facilities<br>PROC8b - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities<br>PROC9 - Transfer of substance or preparation into small containers (dedicated filling line, including weighing) |
| <b>Environmental release category(ies)</b>  | ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates)<br>ERC6b - Industrial use of reactive processing aids<br>ERC7 - Industrial use of substances in closed systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

#### Section 2 - Operational Conditions and Risk Management Measures

##### Product characteristics

|                                   |                          |
|-----------------------------------|--------------------------|
| <b>Physical State</b>             | Liquid                   |
| <b>pH</b>                         | No information available |
| <b>Water Solubility</b>           | Miscible                 |
| <b>Vapor Pressure</b>             | 97 mbar @ 20 °C          |
| <b>Volatility</b>                 | High                     |
| Covers concentrations up to 100 % |                          |

#### Section 2.1 - Control of environmental exposure

##### Environmental release category(ies)

ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates)  
ERC6b - Industrial use of reactive processing aids

## ERC7 - Industrial use of substances in closed systems

**Specific Environmental Release Category**

ESVOC SpERC 1.1.v1

**Control of environmental exposure**

Readily biodegradable

Regional use tonnage 1000 t/a

Annual site tonnage 1000 t/a

Fraction of EU tonnage used in region 1%

Fraction of regional tonnage used locally 1%

**Other operational conditions of use affecting environmental exposure**

Emission days

100

Release fraction to air from process (initial release prior to RMM)

**ERC6a** = 5%**ERC6b** = 0.1%**ERC7** = 5%

Release fraction to wastewater from process (initial release prior to RMM)

**ERC6a** = 2%**ERC6b** = 5%**ERC7** = 5%

Release fraction to soil from process (initial release prior to RMM)

**ERC6a** = 0.1%**ERC6b** = 0.025%**ERC7** = 5%

Remarks

ERC defaults

**Conditions and measures related to municipal sewage treatment plant**

Assumed on-site sewage treatment plant flow

2000m3/d

Sludge treatment

Controlled application to agricultural soil.

**Waste management**

Air

**ERC6a** = 500 kg/day**ERC6b** = 10 kg/day**ERC7** = 500 kg/day

ERC release factor

Water

**ERC6a** = 200 kg/day**ERC6b** = 500 kg/day**ERC7** = 500 kg/day

ERC release factor

Soil

**ERC6a** = 0.1%**ERC6b** = 0.025%**ERC7** = 5%

ERC release factor

**Regional exposure levels and environmental concentrations**

Regional exposure for the use has been modelled using EUSES 2.1. No significant PEC values are indicated for the regional scale even under the conservative assumptions of the Tier 2 EUSES assessment.

**Section 2.2 - Control of worker exposure****General information on risk management related to physicochemical hazard**

Remove all sources of ignition. Take precautionary measures against static charges. Use only non-sparking tools. Keep equipment under negative pressure. Segregate work area and mark with appropriate signs in accordance with local/regional/national legislation.

**General information on exposure estimation**

Used in manufacturing processes which are either closed, continuous processes, or closed batch processes and in batch synthesis where some opportunity for exposure may arise. Respiratory protection is not required except for certain critical activities where respiratory protective equipment is used, for example, cleaning tanks or reactors. Local exhaust ventilation (LEV) usually required for indoor industrial use. Measured dermal exposure data are not available.

**Control of worker exposure**

|                                                                                        |                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Process category(ies)                                                                  | PROC1 - Use in closed process, no likelihood of exposure                                                                                                                                                                                                  |
| Covers concentrations up to                                                            | 100%                                                                                                                                                                                                                                                      |
| Amounts used                                                                           | >1000 t/y                                                                                                                                                                                                                                                 |
| Exposure duration                                                                      | Avoid carrying out operation for more than 8h                                                                                                                                                                                                             |
| Use frequency                                                                          | 220 days per year                                                                                                                                                                                                                                         |
| Indoor/Outdoor use                                                                     | Outdoor                                                                                                                                                                                                                                                   |
| Assumes process temperature up to                                                      | <=40°C                                                                                                                                                                                                                                                    |
| Organisational measures to prevent /limit releases, dispersion and exposure            | Handle substance within a closed system Use of closed transfers of liquids from storage to production equipment (e.g. metered piped or pumped additions) Sample via a closed loop or other system to avoid exposure                                       |
| Technical conditions and measures to control dispersion from source towards the worker | Undertake operation under enclosed conditions                                                                                                                                                                                                             |
| Conditions and measures related to personal protection, hygiene and health evaluation  | Use eye protection according to EN 166, designed to protect against liquid splashes<br>Wear chemically resistant gloves (tested to EN374) in combination with specific activity training                                                                  |
| Additional good practice advice beyond the REACH Chemical Safety Report                | Workers involved in production, handling, sampling and transfer of materials are well-trained in these procedures as well as good industrial hygiene practices<br>-----                                                                                   |
| Process category(ies)                                                                  | PROC2 - Use in closed, continuous process with occasional controlled exposure                                                                                                                                                                             |
| Covers concentrations up to                                                            | 100%                                                                                                                                                                                                                                                      |
| Exposure duration                                                                      | Avoid carrying out operation for more than 8h                                                                                                                                                                                                             |
| Indoor/Outdoor use                                                                     | Outdoor                                                                                                                                                                                                                                                   |
| Assumes process temperature up to                                                      | <=40°C                                                                                                                                                                                                                                                    |
| Conditions and measures related to personal protection, hygiene and health evaluation  | Wear a respirator providing a minimum efficiency of 90% (APF 10) Use eye protection according to EN 166, designed to protect against liquid splashes Wear chemically resistant gloves (tested to EN374) in combination with specific activity training    |
| Additional good practice advice beyond the REACH Chemical Safety Report                | Workers involved in production, handling, sampling and transfer of materials are well-trained in these procedures as well as good industrial hygiene practices<br>-----                                                                                   |
| Process category(ies)                                                                  | PROC3 - Use in closed batch process (synthesis or formulation)                                                                                                                                                                                            |
| Covers concentrations up to                                                            | 100%                                                                                                                                                                                                                                                      |
| Exposure duration                                                                      | < 1 hour(s)                                                                                                                                                                                                                                               |
| Indoor/Outdoor use                                                                     | Outdoor                                                                                                                                                                                                                                                   |
| Assumes process temperature up to                                                      | <=40°C                                                                                                                                                                                                                                                    |
| Conditions and measures related to personal protection, hygiene and health evaluation  | Use eye protection according to EN 166, designed to protect against liquid splashes Wear chemically resistant gloves (tested to EN374) in combination with specific activity training<br>Wear a respirator providing a minimum efficiency of 90% (APF 10) |
| Additional good practice advice beyond the REACH Chemical Safety Report                | Workers involved in production, handling, sampling and transfer of materials are well-trained in these procedures as well as good industrial hygiene practices<br>-----                                                                                   |
| Process category(ies)                                                                  | PROC4 - Use in batch and other process (synthesis) where opportunity for exposure arises                                                                                                                                                                  |
| Covers concentrations up to                                                            | 100%                                                                                                                                                                                                                                                      |
| Exposure duration                                                                      | Avoid carrying out activities involving exposure for more than 1 hour                                                                                                                                                                                     |
| Indoor/Outdoor use                                                                     | Outdoor                                                                                                                                                                                                                                                   |
| Assumes process temperature up to                                                      | <=40°C                                                                                                                                                                                                                                                    |
| Conditions and measures related to personal protection, hygiene and health evaluation  | Use eye protection according to EN 166, designed to protect against liquid splashes Wear a respirator providing a minimum efficiency of 90% (APF 10) Wear chemically resistant gloves (tested to EN374) in combination with specific activity training    |
| Additional good practice advice beyond the REACH Chemical Safety Report                | Workers involved in production, handling, sampling and transfer of materials are well-trained in these procedures as well as good industrial hygiene practices<br>-----                                                                                   |
| Process category(ies)                                                                  | PROC8a - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at non dedicated facilities                                                                                                                         |
| Covers concentrations up to                                                            | 100%                                                                                                                                                                                                                                                      |
| Exposure duration                                                                      | < 1 hour(s)                                                                                                                                                                                                                                               |
| Indoor/Outdoor use                                                                     | Outdoor                                                                                                                                                                                                                                                   |
| Assumes process temperature up to                                                      | <=40°C                                                                                                                                                                                                                                                    |

|                                                                                       |                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Organisational measures to prevent /limit releases, dispersion and exposure           | Avoid carrying out operation for more than 1 hour<br>Ensure operation is undertaken outdoors                                                                                                                                                                 |
| Conditions and measures related to personal protection, hygiene and health evaluation | Use eye protection according to EN 166, designed to protect against liquid splashes<br>Wear a respirator providing a minimum efficiency of 95% (APF 20)<br>Wear chemically resistant gloves (tested to EN374) in combination with specific activity training |
| Additional good practice advice beyond the REACH Chemical Safety Report               | Assumes a good basic standard of occupational hygiene is implemented<br>-----                                                                                                                                                                                |
| Process category(ies)                                                                 | PROC8b - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities                                                                                                                                |
| Covers concentrations up to                                                           | 100%                                                                                                                                                                                                                                                         |
| Exposure duration                                                                     | Avoid carrying out activities involving exposure for more than 1 hour                                                                                                                                                                                        |
| Indoor/Outdoor use                                                                    | Outdoor                                                                                                                                                                                                                                                      |
| Assumes process temperature up to                                                     | <=40°C                                                                                                                                                                                                                                                       |
| Conditions and measures related to personal protection, hygiene and health evaluation | Use eye protection according to EN 166, designed to protect against liquid splashes<br>Wear chemically resistant gloves (tested to EN374) in combination with specific activity training                                                                     |
| Additional good practice advice beyond the REACH Chemical Safety Report               | Assumes a good basic standard of occupational hygiene is implemented<br>-----                                                                                                                                                                                |
| Process category(ies)                                                                 | PROC9 - Transfer of substance or preparation into small containers (dedicated filling line, including weighing)                                                                                                                                              |
| Covers concentrations up to                                                           | 100%                                                                                                                                                                                                                                                         |
| Exposure duration                                                                     | Avoid carrying out operation for more than 8h                                                                                                                                                                                                                |
| Indoor/Outdoor use                                                                    | Indoor use                                                                                                                                                                                                                                                   |
| Assumes process temperature up to                                                     | <=40°C                                                                                                                                                                                                                                                       |
| Minimum room ventilation rate for handling/application (air changes per hour)         | 1-3                                                                                                                                                                                                                                                          |
| Organisational measures to prevent /limit releases, dispersion and exposure           | Fill containers/cans at dedicated fill points supplied with local extract ventilation                                                                                                                                                                        |
| Conditions and measures related to personal protection, hygiene and health evaluation | Use eye protection according to EN 166, designed to protect against liquid splashes<br>Wear chemically resistant gloves (tested to EN374) in combination with specific activity training<br>Wear a respirator providing a minimum efficiency of 90%<br>----- |
| <b>Control of consumer exposure</b>                                                   | Not intended for consumer use                                                                                                                                                                                                                                |

### Section 3 - Exposure estimation

#### Environment

##### Environmental release category(ies)

ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates)

ERC6b - Industrial use of reactive processing aids

ERC7 - Industrial use of substances in closed systems

**Specific Environmental Release Category** - ESVOC SpERC 1.1.v1

**Predicted No Effect Concentration (PNEC)** - See values below

|                                           |                                 |                                          |               |
|-------------------------------------------|---------------------------------|------------------------------------------|---------------|
| <b>Fresh water</b>                        | 10 mg/l                         | <b>Marine water</b>                      | 1 mg/l        |
| <b>Fresh water sediment</b>               | 45 mg/kg dw                     | <b>Marine water sediment</b>             | 4.5 mg/kg dw  |
| <b>Water Intermittent</b>                 | 10 mg/l                         | <b>Soil (Agriculture)</b>                | 3.02 mg/kg dw |
| <b>Microorganisms in sewage treatment</b> | 32 mg/l                         |                                          |               |
| <b>Environment</b>                        | <b>Predicted exposure level</b> | <b>Risk characterization ratio (RCR)</b> |               |
| <b>Freshwater</b>                         | ERC6a = 1.24 mg/l               | 0.122                                    |               |
|                                           | ERC6b = 3.11 mg/l               | 0.304                                    |               |



|                     |                                                                          |                         |
|---------------------|--------------------------------------------------------------------------|-------------------------|
|                     | ERC7 = 3.11 mg/l                                                         | 0.304                   |
| Marine water        | ERC6a = 0.124 mg/l<br>ERC6b = 0.311 mg/l<br>ERC7 = 0.311 mg/l            | 0.122<br>0.304<br>0.304 |
| Freshwater sediment | ERC6a = 5.48 mg/kg dw<br>ERC6b = 13.7 mg/kg dw<br>ERC7 = 13.7 mg/kg dw   | 0.122<br>0.304<br>0.304 |
| Marine sediment     | ERC6a = 0.548 mg/kg dw<br>ERC6b = 1.37 mg/kg dw<br>ERC7 = 1.37 mg/kg dw  | 0.122<br>0.304<br>0.304 |
| Soil                | ERC6a = 0.199 mg/kg dw<br>ERC6b = 0.485 mg/kg dw<br>ERC7 = 0.49 mg/kg dw | 0.657<br>0.16<br>0.162  |
| Municipal STP       | ERC6a = 12.4 mg/l<br>ERC6b = 31.1 mg/l<br>ERC7 = 31.1 mg/l               | 0.388<br>0.976<br>0.97  |

Calculation method - EUSES 2.1

#### Remarks

All RCRs are less than 1, it is considered that safe use has been demonstrated. Atmospheric contamination is minimal, and as there is no indicated effects on plants or animals from the atmosphere, no PNEC/ RCR is derived.

#### Health

Derived No Effect Level (DNEL) - See table for values

| Route of exposure | Acute effects (local)               | Acute effects (systemic)            | Chronic effects (local)             | Chronic effects (systemic)          |
|-------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| Oral              |                                     |                                     |                                     | 32.2 mg/kg bw/day                   |
| Dermal            |                                     |                                     |                                     | 40.6 ppm                            |
| Inhalation        | 40.6 ppm<br>(68 mg/m <sup>3</sup> ) | 40.6 ppm<br>(68 mg/m <sup>3</sup> ) | 40.6 ppm<br>(68 mg/m <sup>3</sup> ) | 40.6 ppm<br>(68 mg/m <sup>3</sup> ) |

| Process category(ies)                                                                                                             | Exposure route      | Predicted exposure level | Risk characterization ratio (RCR) |
|-----------------------------------------------------------------------------------------------------------------------------------|---------------------|--------------------------|-----------------------------------|
| PROC1 - Use in closed process, no likelihood of exposure                                                                          | Worker - inhalative | 0.012 mg/m <sup>3</sup>  | <0.01                             |
|                                                                                                                                   | Worker - dermal     | 0.343 mg/kg bw/day       | 0.011                             |
| PROC2 - Use in closed, continuous process with occasional controlled exposure                                                     | Worker - inhalative | 12.0 mg/m <sup>3</sup>   | 0.179                             |
|                                                                                                                                   | Worker - dermal     | 1.37 mg/kg bw/day        | 0.043                             |
| PROC3 - Use in closed batch process (synthesis or formulation)                                                                    | Worker - inhalative | 29.9 mg/m <sup>3</sup>   | 0.447                             |
|                                                                                                                                   | Worker - dermal     | 0.343 mg/kg bw/day       | 0.011                             |
| PROC4 - Use in batch and other process (synthesis) where opportunity for exposure arises                                          | Worker - inhalative | 24.0 mg/m <sup>3</sup>   | 0.357                             |
|                                                                                                                                   | Worker - dermal     | 6.86 mg/kg bw/day        | 0.214                             |
| PROC8a - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at non dedicated facilities | Worker - inhalative | 60.0 mg/m <sup>3</sup>   | 0.894                             |
|                                                                                                                                   | Worker - dermal     | 12.0 mg/kg bw/day        | 0.429                             |
| PROC8b - Transfer of substance or preparation (charging/discharging) from/to                                                      | Worker - inhalative | 60.0 mg/m <sup>3</sup>   | 0.894                             |

vessels/large containers at dedicated facilities

|                 |                   |       |
|-----------------|-------------------|-------|
| Worker - dermal | 6.86 mg/kg bw/day | 0.214 |
|-----------------|-------------------|-------|

PROC9 - Transfer of substance or preparation into small containers (dedicated filling line, including weighing)

|                     |                         |       |
|---------------------|-------------------------|-------|
| Worker - inhalative | 0.855 mg/m <sup>3</sup> | 0.013 |
|---------------------|-------------------------|-------|

|                 |                   |       |
|-----------------|-------------------|-------|
| Worker - dermal | 6.86 mg/kg bw/day | 0.021 |
|-----------------|-------------------|-------|

**Calculation method**

Used ECETOC TRA model

**Remarks**

Predicted exposures are not expected to exceed the DN(M)EL when the risk management measures/operational conditions outlined in section 2 are implemented

## Section 4 - Guidance to check compliance with the exposure scenario

Used EUSES model

Used ECETOC TRA model

Further details on scaling and control technologies are provided in SpERC factsheet (<http://cefic.org/en/reach-for-industries-libraries.html>)

Predicted exposures are not expected to exceed the applicable exposure limits (given in section 8 of the SDS) when the operational conditions/risk management measures given in section 2 are implemented

ECHA guidance for downstream users

## Annex to the Safety Data Sheet according to Regulation (EC) No 1907/2006 [REACH]

### Acetonitrile - Exposure Scenarios

|                   |                                                    |                    |
|-------------------|----------------------------------------------------|--------------------|
| CAS No<br>75-05-8 | REACH registration number<br>01-2119471307-38-xxxx | EC No<br>203-726-8 |
|-------------------|----------------------------------------------------|--------------------|

#### Exposure scenario

#### ES3 Pharmaceutical, fine chemical and active substance manufacture uses of acetonitrile - ES3-M3 ACETONITRILE

#### Section 1 - Identification of the use

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Main user group                      | Industrial uses: Uses of substances as such or in preparations at industrial sites                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Type                                 | Worker                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Processes, tasks, activities covered | Manufacture or use as an intermediate or process chemical or extraction agent. Loading (including marine vessel/barge, rail/road car and IBC loading) and repacking (including drums and small packs) of substance, including its sampling, storage, unloading distribution and associated laboratory activities                                                                                                                                                                                                                                                                                                    |
| Sector(s) of use                     | SU9 - Manufacture of fine chemicals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Process category(ies)                | PROC1 - Use in closed process, no likelihood of exposure<br>PROC2 - Use in closed, continuous process with occasional controlled exposure<br>PROC3 - Use in closed batch process (synthesis or formulation)<br>PROC4 - Use in batch and other process (synthesis) where opportunity for exposure arises<br>PROC8a - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at non dedicated facilities<br>PROC8b - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities<br>PROC15 - Use as laboratory reagent |
| Environmental release category(ies)  | ERC4 - Industrial use of processing aids in processes and products, not becoming part of articles<br>ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates)                                                                                                                                                                                                                                                                                                                                                                                                                    |

#### Section 2 - Operational Conditions and Risk Management Measures

##### Product characteristics

|                  |                          |
|------------------|--------------------------|
| Physical State   | Liquid                   |
| pH               | No information available |
| Water Solubility | Miscible                 |
| Vapor Pressure   | 97 mbar @ 20 °C          |
| Volatility       | High                     |

Covers concentrations up to 100 %

#### Section 2.1 - Control of environmental exposure

##### Environmental release category(ies)

ERC4 - Industrial use of processing aids in processes and products, not becoming part of articles  
ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates)

## Specific Environmental Release Category

ESVOC SpERC 1.1.v1

### Control of environmental exposure

Readily biodegradable

Regional use tonnage 1000 t/a

Annual site tonnage 500 t/a

Fraction of EU tonnage used in region 1%

Fraction of regional tonnage used locally 0.1%

### Other operational conditions of use affecting environmental exposure

Emission days 200

Release fraction to air from process (initial release prior to RMM) **ERC4** = 100%  
**ERC6a** = 5%

Release fraction to wastewater from process (initial release prior to RMM) **ERC4** = 100%  
**ERC6a** = 2%

Release fraction to soil from process (initial release prior to RMM) **ERC4** = 5%  
**ERC6a** = 0.1%

Remarks ERC defaults

### Conditions and measures related to municipal sewage treatment plant

Assumed on-site sewage treatment plant flow 2000m3/d

Sludge treatment Controlled application to agricultural soil.

### Waste management

Air **ERC4** = 2,500 kg/day  
**ERC6a** = 125kg/day  
ERC release factor  
Water **ERC4** = 2,500 kg/day  
**ERC6a** = 50 kg/day  
ERC release factor  
Soil **ERC4** = 5%  
**ERC6a** = 0.1%  
ERC release factor

### Regional exposure levels and environmental concentrations

Regional exposure for the use has been modelled using EUSES 2.1. No significant PEC values are indicated for the regional scale even under the conservative assumptions of the Tier 2 EUSES assessment.

## Section 2.2 - Control of worker exposure

### General information on risk management related to physicochemical hazard

Remove all sources of ignition. Take precautionary measures against static charges. Use only non-sparking tools. Keep equipment under negative pressure. Segregate work area and mark with appropriate signs in accordance with local/regional/national legislation.

### General information on exposure estimation

Used in manufacturing processes which are either closed, continuous processes, or closed batch processes and in batch synthesis where some opportunity for exposure may arise. Respiratory protection is not required except for certain critical activities where respiratory protective equipment is used, for example, cleaning tanks or reactors. Local exhaust ventilation (LEV) usually required for indoor industrial use. Measured dermal exposure data are not available.

### Control of worker exposure

Process category(ies) PROC1 - Use in closed process, no likelihood of exposure  
Covers concentrations up to 100%  
Amounts used >1000 t/y  
Exposure duration Avoid carrying out operation for more than 8h

|                                                                                        |                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use frequency                                                                          | 220 days per year                                                                                                                                                                                                                                         |
| Indoor/Outdoor use                                                                     | Outdoor                                                                                                                                                                                                                                                   |
| Assumes process temperature up to                                                      | <=40°C                                                                                                                                                                                                                                                    |
| Organisational measures to prevent /limit releases, dispersion and exposure            | Handle substance within a closed system Use of closed transfers of liquids from storage to production equipment (e.g. metered piped or pumped additions) Sample via a closed loop or other system to avoid exposure                                       |
| Technical conditions and measures to control dispersion from source towards the worker | Undertake operation under enclosed conditions                                                                                                                                                                                                             |
| Conditions and measures related to personal protection, hygiene and health evaluation  | Use eye protection according to EN 166, designed to protect against liquid splashes<br>Wear chemically resistant gloves (tested to EN374) in combination with specific activity training                                                                  |
| Additional good practice advice beyond the REACH Chemical Safety Report                | Workers involved in production, handling, sampling and transfer of materials are well-trained in these procedures as well as good industrial hygiene practices<br>-----                                                                                   |
| Process category(ies)                                                                  | PROC2 - Use in closed, continuous process with occasional controlled exposure                                                                                                                                                                             |
| Covers concentrations up to                                                            | 100%                                                                                                                                                                                                                                                      |
| Exposure duration                                                                      | Avoid carrying out operation for more than 8h                                                                                                                                                                                                             |
| Indoor/Outdoor use                                                                     | Outdoor                                                                                                                                                                                                                                                   |
| Assumes process temperature up to                                                      | <=40°C                                                                                                                                                                                                                                                    |
| Conditions and measures related to personal protection, hygiene and health evaluation  | Wear a respirator providing a minimum efficiency of 90% (APF 10) Use eye protection according to EN 166, designed to protect against liquid splashes Wear chemically resistant gloves (tested to EN374) in combination with specific activity training    |
| Additional good practice advice beyond the REACH Chemical Safety Report                | Workers involved in production, handling, sampling and transfer of materials are well-trained in these procedures as well as good industrial hygiene practices<br>-----                                                                                   |
| Process category(ies)                                                                  | PROC3 - Use in closed batch process (synthesis or formulation)                                                                                                                                                                                            |
| Covers concentrations up to                                                            | 100%                                                                                                                                                                                                                                                      |
| Exposure duration                                                                      | < 1 hour(s)                                                                                                                                                                                                                                               |
| Indoor/Outdoor use                                                                     | Outdoor                                                                                                                                                                                                                                                   |
| Assumes process temperature up to                                                      | <=40°C                                                                                                                                                                                                                                                    |
| Conditions and measures related to personal protection, hygiene and health evaluation  | Use eye protection according to EN 166, designed to protect against liquid splashes Wear chemically resistant gloves (tested to EN374) in combination with specific activity training<br>Wear a respirator providing a minimum efficiency of 90% (APF 10) |
| Additional good practice advice beyond the REACH Chemical Safety Report                | Workers involved in production, handling, sampling and transfer of materials are well-trained in these procedures as well as good industrial hygiene practices<br>-----                                                                                   |
| Process category(ies)                                                                  | PROC4 - Use in batch and other process (synthesis) where opportunity for exposure arises                                                                                                                                                                  |
| Covers concentrations up to                                                            | 100%                                                                                                                                                                                                                                                      |
| Exposure duration                                                                      | Avoid carrying out activities involving exposure for more than 1 hour                                                                                                                                                                                     |
| Indoor/Outdoor use                                                                     | Outdoor                                                                                                                                                                                                                                                   |
| Assumes process temperature up to                                                      | <=40°C                                                                                                                                                                                                                                                    |
| Conditions and measures related to personal protection, hygiene and health evaluation  | Use eye protection according to EN 166, designed to protect against liquid splashes Wear a respirator providing a minimum efficiency of 90% (APF 10) Wear chemically resistant gloves (tested to EN374) in combination with specific activity training    |
| Additional good practice advice beyond the REACH Chemical Safety Report                | Workers involved in production, handling, sampling and transfer of materials are well-trained in these procedures as well as good industrial hygiene practices<br>-----                                                                                   |
| Process category(ies)                                                                  | PROC8a - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at non dedicated facilities                                                                                                                         |
| Covers concentrations up to                                                            | 100%                                                                                                                                                                                                                                                      |
| Exposure duration                                                                      | < 1 hour(s)                                                                                                                                                                                                                                               |
| Indoor/Outdoor use                                                                     | Outdoor                                                                                                                                                                                                                                                   |
| Assumes process temperature up to                                                      | <=40°C                                                                                                                                                                                                                                                    |
| Organisational measures to prevent /limit releases, dispersion and exposure            | Avoid carrying out operation for more than 1 hour<br>Ensure operation is undertaken outdoors                                                                                                                                                              |
| Conditions and measures related to personal protection, hygiene and health evaluation  | Use eye protection according to EN 166, designed to protect against liquid splashes<br>Wear a respirator providing a minimum efficiency of 95% (APF 20)<br>Wear chemically resistant gloves (tested to EN374) in combination with specific activity       |

|                                                                                       |                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                       | training                                                                                                                                                                              |
| Additional good practice advice beyond the REACH Chemical Safety Report               | Assumes a good basic standard of occupational hygiene is implemented                                                                                                                  |
| -----                                                                                 |                                                                                                                                                                                       |
| Process category(ies)                                                                 | PROC8b - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities                                                         |
| Covers concentrations up to                                                           | 100%                                                                                                                                                                                  |
| Exposure duration                                                                     | Avoid carrying out activities involving exposure for more than 1 hour                                                                                                                 |
| Indoor/Outdoor use                                                                    | Outdoor                                                                                                                                                                               |
| Assumes process temperature up to                                                     | <=40°C                                                                                                                                                                                |
| Conditions and measures related to personal protection, hygiene and health evaluation | Use eye protection according to EN 166, designed to protect against liquid splashes Wear chemically resistant gloves (tested to EN374) in combination with specific activity training |
| Additional good practice advice beyond the REACH Chemical Safety Report               | Assumes a good basic standard of occupational hygiene is implemented                                                                                                                  |

|                                                                                       |                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Process category(ies)                                                                 | PROC15 - Use as laboratory reagent                                                                                                                                                                                                            |
| Covers concentrations up to                                                           | 100%                                                                                                                                                                                                                                          |
| Exposure duration                                                                     | Avoid carrying out operation for more than 8h                                                                                                                                                                                                 |
| Indoor/Outdoor use                                                                    | Indoor use                                                                                                                                                                                                                                    |
| Assumes process temperature up to                                                     | <=40°C                                                                                                                                                                                                                                        |
| Conditions and measures related to personal protection, hygiene and health evaluation | Use eye protection according to EN 166, designed to protect against liquid splashes Wear chemically resistant gloves (tested to EN374) in combination with specific activity training Wear a respirator providing a minimum efficiency of 90% |
| Additional good practice advice beyond the REACH Chemical Safety Report               | Workers involved in production, handling, sampling and transfer of materials are well-trained in these procedures as well as good industrial hygiene practices                                                                                |

**Control of consumer exposure** Not intended for consumer use

### Section 3 - Exposure estimation

#### Environment

##### Environmental release category(ies)

ERC4 - Industrial use of processing aids in processes and products, not becoming part of articles

ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates)

**Specific Environmental Release Category** - ESVOC SpERC 1.1.v1

**Predicted No Effect Concentration (PNEC)** - See values below

|                                           |             |                              |               |
|-------------------------------------------|-------------|------------------------------|---------------|
| <b>Fresh water</b>                        | 10 mg/l     | <b>Marine water</b>          | 1 mg/l        |
| <b>Fresh water sediment</b>               | 45 mg/kg dw | <b>Marine water sediment</b> | 4.5 mg/kg dw  |
| <b>Water Intermittent</b>                 | 10 mg/l     | <b>Soil (Agriculture)</b>    | 3.02 mg/kg dw |
| <b>Microorganisms in sewage treatment</b> | 32 mg/l     |                              |               |

| <u>Environment</u>  | <u>Predicted exposure level</u>                | <u>Risk characterization ratio (RCR)</u> |
|---------------------|------------------------------------------------|------------------------------------------|
| Freshwater          | ERC4a = 3.21 mg/l<br>ERC6a = 0.311 mg/l        | 0.315<br>0.0305                          |
| Marine water        | ERC4 = 0.321 mg/l<br>ERC6a = 0.0311 mg/l       | 0.315<br>0.0305                          |
| Freshwater sediment | ERC4 = 14.2 mg/kg dw<br>ERC6a = 1.37 mg/kg dw  | 0.315<br>0.0305                          |
| Marine sediment     | ERC4 = 1.42 mg/kg dw<br>ERC6a = 0.137 mg/kg dw | 0.315<br>0.0305                          |

|               |                                                                    |                 |
|---------------|--------------------------------------------------------------------|-----------------|
| Soil          | ERC4 = 2.47 mg/kg dw<br>ERC6a = 0.0509 mg/kg dw                    | 0.818<br>0.0168 |
| Municipal STP | ERC4 = 32 mg/l<br>ERC6a = 3.11 mg/l                                | 1<br>0.097      |
| Air           | ERC4 = 0.381 mg/m <sup>3</sup><br>ERC6a = 0.0191 mg/m <sup>3</sup> |                 |

Calculation method - EUSES 2.1

#### Remarks

All RCRs are less than 1, it is considered that safe use has been demonstrated. Atmospheric contamination is minimal, and as there is no indicated effects on plants or animals from the atmosphere, no PNEC/ RCR is derived.

#### Health

Derived No Effect Level (DNEL) - See table for values

| Route of exposure | Acute effects (local)               | Acute effects (systemic)            | Chronic effects (local)             | Chronic effects (systemic)          |
|-------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| Oral              |                                     |                                     |                                     | 32.2 mg/kg bw/day                   |
| Dermal            |                                     |                                     |                                     | 40.6 ppm                            |
| Inhalation        | 40.6 ppm<br>(68 mg/m <sup>3</sup> ) | 40.6 ppm<br>(68 mg/m <sup>3</sup> ) | 40.6 ppm<br>(68 mg/m <sup>3</sup> ) | 40.6 ppm<br>(68 mg/m <sup>3</sup> ) |

| Process category(ies)                                                                                                             | Exposure route      | Predicted exposure level | Risk characterization ratio (RCR) |
|-----------------------------------------------------------------------------------------------------------------------------------|---------------------|--------------------------|-----------------------------------|
| PROC1 - Use in closed process, no likelihood of exposure                                                                          | Worker - inhalative | 0.012 mg/m <sup>3</sup>  | <0.01                             |
|                                                                                                                                   | Worker - dermal     | 0.343 mg/kg bw/day       | 0.011                             |
| PROC2 - Use in closed, continuous process with occasional controlled exposure                                                     | Worker - inhalative | 12.0 mg/m <sup>3</sup>   | 0.179                             |
|                                                                                                                                   | Worker - dermal     | 1.37 mg/kg bw/day        | 0.043                             |
| PROC3 - Use in closed batch process (synthesis or formulation)                                                                    | Worker - inhalative | 29.9 mg/m <sup>3</sup>   | 0.447                             |
|                                                                                                                                   | Worker - dermal     | 0.343 mg/kg bw/day       | 0.011                             |
| PROC4 - Use in batch and other process (synthesis) where opportunity for exposure arises                                          | Worker - inhalative | 24.0 mg/m <sup>3</sup>   | 0.357                             |
|                                                                                                                                   | Worker - dermal     | 6.86 mg/kg bw/day        | 0.214                             |
| PROC8a - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at non dedicated facilities | Worker - inhalative | 60.0 mg/m <sup>3</sup>   | 0.894                             |
|                                                                                                                                   | Worker - dermal     | 12.0 mg/kg bw/day        | 0.429                             |
| PROC8b - Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities     | Worker - inhalative | 60.0 mg/m <sup>3</sup>   | 0.894                             |
|                                                                                                                                   | Worker - dermal     | 6.86 mg/kg bw/day        | 0.214                             |
| PROC15 - Use as laboratory reagent                                                                                                | Worker - inhalative | 1.71 mg/m <sup>3</sup>   | 0.026                             |
|                                                                                                                                   | Worker - dermal     | 0.0343 mg/kg bw/day      | 0.001                             |

Calculation method Used ECETOC TRA model

#### Remarks

Predicted exposures are not expected to exceed the DN(M)EL when the risk management measures/operational conditions outlined in section 2 are implemented

## Section 4 - Guidance to check compliance with the exposure scenario

Used EUSES model

Used ECETOC TRA model

Further details on scaling and control technologies are provided in SpERC factsheet  
(<http://cefic.org/en/reach-for-industries-libraries.html>)

Predicted exposures are not expected to exceed the applicable exposure limits (given in section 8 of the SDS) when the operational conditions/risk management measures given in section 2 are implemented

ECHA guidance for downstream users



## Annex to the Safety Data Sheet according to Regulation (EC) No 1907/2006 [REACH]

### Acetonitrile - Exposure Scenarios

|                   |                                                    |                    |
|-------------------|----------------------------------------------------|--------------------|
| CAS No<br>75-05-8 | REACH registration number<br>01-2119471307-38-xxxx | EC No<br>203-726-8 |
|-------------------|----------------------------------------------------|--------------------|

#### Exposure scenario

#### ES4 Repackaging of Acetonitrile - ES4-F1 ACETONITRILE

#### Section 1 - Identification of the use

|                                             |                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Main user group</b>                      | Industrial uses: Uses of substances as such or in preparations at industrial sites                                                                                                                                                                                                                                      |
| <b>Type</b>                                 | Worker                                                                                                                                                                                                                                                                                                                  |
| <b>Processes, tasks, activities covered</b> | Formulation, packing and re-packing of the substance and its mixtures in batch or continuous operations, including storage, materials transfers, mixing, tableting, compression, pelletisation, extrusion, large and small scale packing, sampling, maintenance and associated laboratory activities.                   |
| <b>Sector(s) of use</b>                     | SU10 - Formulation [mixing] of preparations and/or re-packaging (excluding alloys)                                                                                                                                                                                                                                      |
| <b>Process category(ies)</b>                | PROC3 - Use in closed batch process (synthesis or formulation)<br>PROC5 - Mixing or blending in batch processes for formulation of preparations and articles (multistage and/or significant contact)<br>PROC9 - Transfer of substance or preparation into small containers (dedicated filling line, including weighing) |
| <b>Environmental release category(ies)</b>  | ERC2 - Formulation of preparations (mixtures)                                                                                                                                                                                                                                                                           |

#### Section 2 - Operational Conditions and Risk Management Measures

##### Product characteristics

|                                   |                          |
|-----------------------------------|--------------------------|
| <b>Physical State</b>             | Liquid                   |
| <b>pH</b>                         | No information available |
| <b>Water Solubility</b>           | Miscible                 |
| <b>Vapor Pressure</b>             | 97 mbar @ 20 °C          |
| <b>Volatility</b>                 | High                     |
| Covers concentrations up to 100 % |                          |

#### Section 2.1 - Control of environmental exposure

**Environmental release category(ies)**  
ERC2 - Formulation of preparations (mixtures)

**Specific Environmental Release Category**  
ESVOC SpERC 1.1.v1

**Control of environmental exposure**  
Readily biodegradable  
Annual site tonnage 5 t/a  
Fraction of EU tonnage used in region 1%  
Fraction of regional tonnage used locally 1%

**Other operational conditions of use affecting environmental exposure**

|                                                                            |              |
|----------------------------------------------------------------------------|--------------|
| Emission days                                                              | 20           |
| Release fraction to air from process (initial release prior to RMM)        | 2.5%         |
| Release fraction to wastewater from process (initial release prior to RMM) | 2%           |
| Release fraction to soil from process (initial release prior to RMM)       | 0.01%        |
| Remarks                                                                    | ERC defaults |

**Conditions and measures related to municipal sewage treatment plant**

|                                             |                                              |
|---------------------------------------------|----------------------------------------------|
| Assumed on-site sewage treatment plant flow | 2000m3/d                                     |
| Sludge treatment                            | Controlled application to agricultural soil. |

**Waste management**

|       |                              |
|-------|------------------------------|
| Air   | 6.25 kg/d ERC release factor |
| Water | 5 kg/d ERC release factor    |

**Regional exposure levels and environmental concentrations**

Regional exposure for the use has been modelled using EUSES 2.1. No significant PEC values are indicated for the regional scale even under the conservative assumptions of the Tier 2 EUSES assessment.

**Section 2.2 - Control of worker exposure****General information on risk management related to physicochemical hazard**

Remove all sources of ignition. Take precautionary measures against static charges. Use only non-sparking tools. Keep equipment under negative pressure. Segregate work area and mark with appropriate signs in accordance with local/regional/national legislation.

**General information on exposure estimation**

Local exhaust ventilation (LEV) usually required for indoor industrial use. Measured dermal exposure data are not available.

**Control of worker exposure**

|                                                                                       |                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Covers concentrations up to                                                           | 100%                                                                                                                                                                                                                                                   |
| Exposure duration                                                                     | < 8 hour(s)                                                                                                                                                                                                                                            |
| Indoor/Outdoor use                                                                    | Indoor/Outdoor use                                                                                                                                                                                                                                     |
| Assumes process temperature up to                                                     | <=40°C                                                                                                                                                                                                                                                 |
| Organisational measures to prevent /limit releases, dispersion and exposure           | Use engineering controls to keep exposures below the OEL or DNEL                                                                                                                                                                                       |
| Conditions and measures related to personal protection, hygiene and health evaluation | Use eye protection according to EN 166, designed to protect against liquid splashes Wear chemically resistant gloves (tested to EN374) in combination with specific activity training Wear a respirator providing a minimum efficiency of 90% (APF 10) |
| Additional good practice advice beyond the REACH Chemical Safety Report               | Workers involved in production, handling, sampling and transfer of materials are well-trained in these procedures as well as good industrial hygiene practices                                                                                         |

**Control of consumer exposure** Not intended for consumer use

**Section 3 - Exposure estimation****Environment****Environmental release category(ies)**

ERC2 - Formulation of preparations (mixtures)

**Specific Environmental Release Category** - ESVOC SpERC 1.1.v1

**Predicted No Effect Concentration (PNEC)** - See values below

|                                           |             |                              |               |
|-------------------------------------------|-------------|------------------------------|---------------|
| <b>Fresh water</b>                        | 10 mg/l     | <b>Marine water</b>          | 1 mg/l        |
| <b>Fresh water sediment</b>               | 45 mg/kg dw | <b>Marine water sediment</b> | 4.5 mg/kg dw  |
| <b>Water Intermittent</b>                 | 10 mg/l     | <b>Soil (Agriculture)</b>    | 3.02 mg/kg dw |
| <b>Microorganisms in sewage treatment</b> | 32 mg/l     |                              |               |

| <u>Environment</u>                    | <u>Predicted exposure level</u> | <u>Risk characterization ratio (RCR)</u> |
|---------------------------------------|---------------------------------|------------------------------------------|
| Freshwater                            | 0.0311 mg/l                     | $3.04 \times 10^{-3}$                    |
| Marine water                          | $3.11 \times 10^{-3}$ mg/l      | $3.04 \times 10^{-3}$                    |
| Freshwater sediment                   | 0.137 mg/l                      | $3.04 \times 10^{-3}$                    |
| Marine sediment                       | 0.0137 mg/l                     | $3.04 \times 10^{-3}$                    |
| Soil                                  | $4.86 \times 10^{-3}$ mg/kg dw  | $1.61 \times 10^{-3}$                    |
| Municipal STP                         | 0.31 mg/l                       | $9.7 \times 10^{-3}$                     |
| <b>Calculation method</b> - EUSES 2.1 |                                 |                                          |

**Remarks**

All RCRs are less than 1, it is considered that safe use has been demonstrated. Atmospheric contamination is minimal, and as there is no indicated effects on plants or animals from the atmosphere, no PNEC/ RCR is derived.

**Health**

**Derived No Effect Level (DNEL)** - See table for values

| <u>Route of exposure</u> | <u>Acute effects (local)</u>        | <u>Acute effects (systemic)</u>     | <u>Chronic effects (local)</u>      | <u>Chronic effects (systemic)</u>   |
|--------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| Oral                     |                                     |                                     |                                     | 32.2 mg/kg bw/day                   |
| Dermal                   |                                     |                                     |                                     | 40.6 ppm                            |
| Inhalation               | 40.6 ppm<br>(68 mg/m <sup>3</sup> ) | 40.6 ppm<br>(68 mg/m <sup>3</sup> ) | 40.6 ppm<br>(68 mg/m <sup>3</sup> ) | 40.6 ppm<br>(68 mg/m <sup>3</sup> ) |

| <u>Process category(ies)</u>                                                                                                       | <u>Exposure route</u>                    | <u>Predicted exposure level</u> | <u>Risk characterization ratio (RCR)</u> |
|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|---------------------------------|------------------------------------------|
| PROC3 - Use in closed batch process (synthesis or formulation)                                                                     | Worker - inhalative Without LEV          | 42.8 mg/m <sup>3</sup>          | 0.638                                    |
|                                                                                                                                    | Worker - inhalative Without LEV/with RPE | 4.28 mg/m <sup>3</sup>          | 0.064                                    |
|                                                                                                                                    | Worker - inhalative With LEV             | 8.55 mg/m <sup>3</sup>          | 0.128                                    |
|                                                                                                                                    | Worker - dermal                          | 0.343 mg/kg bw/day              | 0.011                                    |
|                                                                                                                                    |                                          |                                 |                                          |
| PROC5 - Mixing or blending in batch processes for formulation of preparations and articles (multistage and/or significant contact) | Worker - inhalative Without LEV          | 171 mg/m <sup>3</sup>           | 2.55                                     |
|                                                                                                                                    | Worker - inhalative Without LEV/with RPE | 17.1 mg/m <sup>3</sup>          | 0.255                                    |
|                                                                                                                                    | Worker - inhalative With LEV             | 34.2 mg/m <sup>3</sup>          | 0.511                                    |
|                                                                                                                                    | Worker - dermal                          | 13.7 mg/kg bw/day               | 0.429                                    |
|                                                                                                                                    |                                          |                                 |                                          |
| PROC9 - Transfer of substance or preparation into small containers (dedicated filling line, including weighing)                    | Worker - inhalative Without LEV          | 171 mg/m <sup>3</sup>           | 2.55                                     |
|                                                                                                                                    | Worker - inhalative Without LEV/with RPE | 17.1 mg/m <sup>3</sup>          | 0.255                                    |
|                                                                                                                                    | Worker - inhalative With LEV             | 34.2 mg/m <sup>3</sup>          | 0.511                                    |
|                                                                                                                                    | Worker - dermal                          | 6.86 mg/kg bw/day               | 0.214                                    |
|                                                                                                                                    |                                          |                                 |                                          |

**Calculation method**

Used ECETOC TRA model

PROC 5 and 9 were found to exceed the DNEL for acute and long-term systemic effects and for acute and long-term local effects when performing tasks indoors without LEV and without respiratory protection

**Remarks**

Predicted exposures are not expected to exceed the DN(M)EL when the risk management measures/operational conditions outlined in section 2 are implemented

## Section 4 - Guidance to check compliance with the exposure scenario

Used EUSES model

Used ECETOC TRA model

Further details on scaling and control technologies are provided in SpERC factsheet

(<http://cefic.org/en/reach-for-industries-libraries.html>)

Predicted exposures are not expected to exceed the applicable exposure limits (given in section 8 of the SDS) when the operational conditions/risk management measures given in section 2 are implemented

ECHA guidance for downstream users

**Annex to the Safety Data Sheet according to Regulation (EC) No 1907/2006 [REACH]****Acetonitrile - Exposure Scenarios**

|                          |                                                           |                           |
|--------------------------|-----------------------------------------------------------|---------------------------|
| <b>CAS No</b><br>75-05-8 | <b>REACH registration number</b><br>01-2119471307-38-xxxx | <b>EC No</b><br>203-726-8 |
|--------------------------|-----------------------------------------------------------|---------------------------|

**Exposure scenario****ES5 Laboratory use of Acetonitrile - ES5-L1 ACETONITRILE****Section 1 - Identification of the use**

|                                             |                                                                                                                                        |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Main user group</b>                      | Industrial uses: Uses of substances as such or in preparations at industrial sites                                                     |
| <b>Type</b>                                 | Worker                                                                                                                                 |
| <b>Processes, tasks, activities covered</b> | Laboratory reagent and solvent involving transfer from larger to small containers and vice versa.                                      |
| <b>Sector(s) of use</b>                     | SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites<br>SU24 - Scientific research and development |
| <b>Process category(ies)</b>                | PROC3 - Use in closed batch process (synthesis or formulation)<br>PROC15 - Use as laboratory reagent                                   |
| <b>Environmental release category(ies)</b>  | ERC8a - Wide dispersive indoor use of processing aids in open systems                                                                  |

**Section 2 - Operational Conditions and Risk Management Measures****Product characteristics**

|                                   |                          |
|-----------------------------------|--------------------------|
| <b>Physical State</b>             | Liquid                   |
| <b>pH</b>                         | No information available |
| <b>Water Solubility</b>           | Miscible                 |
| <b>Vapor Pressure</b>             | 97 mbar @ 20 °C          |
| <b>Volatility</b>                 | High                     |
| Covers concentrations up to 100 % |                          |

**Section 2.1 - Control of environmental exposure****Environmental release category(ies)**

ERC8a - Wide dispersive indoor use of processing aids in open systems

**Specific Environmental Release Category**

ESVOC SpERC 8.17.v1

**Control of environmental exposure**

Readily biodegradable  
Annual site tonnage 2000 t/a  
Fraction of EU tonnage used in region 1%  
Fraction of regional tonnage used locally 0.0005%

**Other operational conditions of use affecting environmental exposure**

Emission days 365  
Release fraction to air from process (initial 50%

release prior to RMM)  
Release fraction to wastewater from process (initial release prior to RMM) 50%

#### Conditions and measures related to municipal sewage treatment plant

Assumed on-site sewage treatment plant flow 2000m<sup>3</sup>/d  
Sludge treatment Controlled application to agricultural soil.

#### Waste management

Air 1.37 kg/day ERC release factor  
Water 1.37 kg/day ERC release factor  
Soil 0.00 kg/d ERC release factor

#### Regional exposure levels and environmental concentrations

Regional exposure for the use has been modelled using EUSES 2.1. No significant PEC values are indicated for the regional scale even under the conservative assumptions of the Tier 2 EUSES assessment.

## Section 2.2 - Control of worker exposure

#### General information on risk management related to physicochemical hazard

Remove all sources of ignition. Take precautionary measures against static charges. Use only non-sparking tools. Keep equipment under negative pressure. Segregate work area and mark with appropriate signs in accordance with local/regional/national legislation.

#### General information on exposure estimation

Local exhaust ventilation (LEV) usually required for indoor industrial use. Measured dermal exposure data are not available.

#### Control of worker exposure

Covers concentrations up to 100%  
Exposure duration < 8 hour(s)  
Indoor/Outdoor use Indoor use  
Assumes process temperature up to ≤40°C  
Technical conditions and measures to control dispersion from source towards the worker Handle in an enclosing hood with exhaust ventilation  
Conditions and measures related to personal protection, hygiene and health evaluation Use eye protection according to EN 166, designed to protect against liquid splashes Wear chemically resistant gloves (tested to EN374) in combination with specific activity training  
Additional good practice advice beyond the REACH Chemical Safety Report Workers involved in production, handling, sampling and transfer of materials are well-trained in these procedures as well as good industrial hygiene practices

**Control of consumer exposure** Not intended for consumer use

## Section 3 - Exposure estimation

#### Environment

##### Environmental release category(ies)

ERC8a - Wide dispersive indoor use of processing aids in open systems

**Specific Environmental Release Category** - ESVOC SpERC 8.17.v1

**Predicted No Effect Concentration (PNEC)** - See values below

|                                           |             |                              |               |
|-------------------------------------------|-------------|------------------------------|---------------|
| <b>Fresh water</b>                        | 10 mg/l     | <b>Marine water</b>          | 1 mg/l        |
| <b>Fresh water sediment</b>               | 45 mg/kg dw | <b>Marine water sediment</b> | 4.5 mg/kg dw  |
| <b>Water Intermittent</b>                 | 10 mg/l     | <b>Soil (Agriculture)</b>    | 3.02 mg/kg dw |
| <b>Microorganisms in sewage treatment</b> | 32 mg/l     |                              |               |

Environment

Predicted exposure level

Risk characterization ratio (RCR)

|                                |                                 |                        |
|--------------------------------|---------------------------------|------------------------|
| Freshwater                     | 0.0112 mg/l                     | 1.1 x10 <sup>-3</sup>  |
| Marine water                   | 1.1 x10 <sup>-3</sup> mg/l      | 1.09 x10 <sup>-3</sup> |
| Freshwater sediment            | 0.0107 mg/kg dw                 | 1.09 x10 <sup>-3</sup> |
| Marine sediment                | 1.06 x10 <sup>-3</sup> mg/kg dw | 1.09 x10 <sup>-3</sup> |
| Soil                           | 1.35 x10 <sup>-3</sup> mg/kg dw | 5.06 x10 <sup>-4</sup> |
| Municipal STP                  | 0.0851 mg/l                     | 2.66 x10 <sup>-3</sup> |
| Air                            | 0.0381 mg/m <sup>3</sup>        |                        |
| Calculation method - EUSES 2.1 |                                 |                        |

**Remarks**

All RCRs are less than 1, it is considered that safe use has been demonstrated. Atmospheric contamination is minimal, and as there is no indicated effects on plants or animals from the atmosphere, no PNEC/ RCR is derived.

**Health**

**Derived No Effect Level (DNEL)** - See table for values

| Route of exposure | Acute effects (local)               | Acute effects (systemic)            | Chronic effects (local)             | Chronic effects (systemic)          |
|-------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| Oral              |                                     |                                     |                                     | 32.2 mg/kg bw/day                   |
| Dermal            |                                     |                                     |                                     | 40.6 ppm                            |
| Inhalation        | 40.6 ppm<br>(68 mg/m <sup>3</sup> ) | 40.6 ppm<br>(68 mg/m <sup>3</sup> ) | 40.6 ppm<br>(68 mg/m <sup>3</sup> ) | 40.6 ppm<br>(68 mg/m <sup>3</sup> ) |

| Process category(ies)                                          | Exposure route                  | Predicted exposure level | Risk characterization ratio (RCR) |
|----------------------------------------------------------------|---------------------------------|--------------------------|-----------------------------------|
| PROC3 - Use in closed batch process (synthesis or formulation) | Worker - inhalative Without LEV | 42.8 mg/m <sup>3</sup>   | 0.638                             |
|                                                                | Worker - inhalative With LEV    | 8.55 mg/m <sup>3</sup>   | 0.128                             |
|                                                                | Worker - dermal                 | 0.343 mg/kg bw/day       | 0.011                             |
| PROC15 - Use as laboratory reagent                             | Worker - inhalative Without LEV | 1.71 mg/m <sup>3</sup>   | 0.255                             |
|                                                                | Worker - inhalative With LEV    | 3.42 mg/m <sup>3</sup>   | 0.051                             |
|                                                                | Worker - dermal                 | 0.0343 mg/kg bw/day      | 0.011                             |

**Calculation method**

Used ECETOC TRA model

**Remarks**

Predicted exposures are not expected to exceed the DN(M)EL when the risk management measures/operational conditions outlined in section 2 are implemented

## Section 4 - Guidance to check compliance with the exposure scenario

Used EUSES model

Used ECETOC TRA model

Further details on scaling and control technologies are provided in SpERC factsheet (<http://cefic.org/en/reach-for-industries-libraries.html>)

Predicted exposures are not expected to exceed the applicable exposure limits (given in section 8 of the SDS) when the operational conditions/risk management measures given in section 2 are implemented

ECHA guidance for downstream users